XIAOQI QIN

Third-Year Undergraduate Student in Mechatronics Engineering

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Research Interests

Video World Models, World-Action Models, Interactive and Long-Horizon Video Generation, Embodied AI.

Education

South China University of Technology Guangzhou, China
B.Eng. in Mechatronics Engineering Sep 2024 - Jun 2028 (Expected)
GPA: 3.90/4.00 | **IELTS:** 7.5

Selected coursework: C++ Programming (100), Probability & Statistics (99), Linear Algebra (97), Calculus I/II (96/96), Introduction to AI (93), Theoretical Mechanics (95), Electric Circuits (90)

Research Experience

Peking University Beijing, China
Research Intern, advised by Prof. Hao Tang Feb 2026 - Present
Map-Conditioned Multi-View Interactive World Modeling *Ongoing; manuscript in preparation*

- Co-developed a multi-view first-person interactive world model for highly dynamic CS:GO scenes by adapting *LingBot-World*, a 14B action-conditioned video DiT; achieved synchronized views with cross-view consistency, a stable world state, complex action-conditioned interaction, and long-horizon generation.
- Designed and trained a residual *Map Memory* conditioning adapter with LoRA for the 14B model; ran and debugged distributed experiments on a **16× NVIDIA H200 GPU cluster**, diagnosing failures across the data, training, and systems stack.
- Built the end-to-end data and evaluation pipeline from raw gameplay recordings to synchronized multi-view clips and *Map Memory* conditions, covering spatial grounding, cross-view consistency, world stability, and long-horizon behavior.

Technical Skills

Programming and frameworks: Python, PyTorch, C++

Generative and world models: Diffusion Transformers (DiT), action-conditioned video generation, multi-view world modeling, conditioning adapters, LoRA fine-tuning

Research engineering: Distributed multi-GPU training, data pipelines, Linux, SSH, tmux, Git, Hugging Face

Honors and Awards

National Scholarship for Undergraduates Sep 2025
Awarded by the Ministry of Education of China to outstanding undergraduate students nationwide; CNY 10,000.

Availability

Guangzhou-based; available for a 6–12 month in-person research appointment.